P-256 factor-base investigation: same-day parallel pass

25 October 2026. All eight planned workstreams have completed a first implementation or mathematical-screening pass. The work uses published P-256 parameters, a secp256k1 automorphism control, P-224 and Curve448 comparisons, and explicit same-field covers. These are exact construction/correctness observations and scoped derivations. No end-to-end ECDLP advantage has been established.

The tracking PR is https://github.com/aburan28/crypto-autoresearcher/pull/2561. The public branch currently contains the plan, handoffs, original F1 package and its cubic-divisor continuation card through c35b9da6a65e019eaa517fc1a5f7774428ff35db. The larger result payload is retained locally while explicit public-disclosure approval is pending after automatic approval review refused the upload. Local isolated engineering checks completed under a recorded custody exception; they are not represented as canonical scientific promotion gates.

What was actually implemented

Track	Concrete result	Remaining decisive obligation
F1 symmetric/coset bases	Exact symmetric circuit; nine domains and 27 complete signed target sets. A new cubic-divisor formulation removes algebraic multiplicities, completes six query sets through depth 3, and retains 21 certificates with 386 independently replayed critical pairs.	Complete implicit extraction and a cost argument in base size/depth. Three depth-4 continuation process stops remain unknown.
F2 auxiliary isogeny bases	Sixteen rational degree-two edges, composite degree 65,536; complete fibres, 16,320 enumerated coordinates, 8,279 accepted target lifts and exact dual checks.	Natural decomposition solving, useful relation density and full cost. Auxiliary smooth group structure does not transfer to P-256.
F3 coefficient-labelled bases	A scalar-blind three-partition hybrid solver, separate exhaustive point-sum control, and distinct useful/zero-Q constructed fixtures. Separate Sage checks cover 72 labels and 6,144 signed sums.	A costed natural useful-dependency solver; the known-witness useful fixture is correctness only. See the coefficient-mod-r clarification below.
F4 divisor stitching	Exact principal-divisor extraction, signed outside-point cancellation and independent rank checks on 29 cases. Five nonrational divisors remain explicitly unresolved.	Natural all-draw yield/collision/rank growth, scalable signed-cycle handling and target attachment.
F5 target representatives	Four explicit maps of degrees 3,5,11,11; symbolic landing and dual identities, 28 point fixtures; complete discriminant certificate gives f_pi=f_E=1.	A matched source/endpoint membership and relation solver paying for discovery and transport. Map existence alone is not cheaper ECDLP.
F6 interval/digit lattices	Twenty-eight rectangle and nine digit cases with exact determinant/norm bounds; one intentionally known-offset extraction control.	Other useful-width constructions and certified ordinary-target multivariate extraction. Failure of this sufficient-bound family is not a general impossibility theorem.
F7 covers	Explicit smooth nonhyperelliptic genus-three and hyperelliptic genus-two covers; subgroup injection through norm after pullback=[2]; fourteen fibre fixtures and a complementary elliptic count.	Arbitrary-class arithmetic/norm and a costed image-specific relation/descent algorithm. A new Cartier-component derivation is separately checked locally.
F8 endomorphism orbits	A genuine signed order-six secp256k1 orbit control: 48 points, 96 constructed rows and exact compression to eight unknowns.	Ordinary-target relation generation, with the same automorphism/compression benefits in the baseline. The fixed symmetry factor is not a new exponent.

Each track's report.md, result.json and report-delivery.pdf are under tracks/f1/ through tracks/f8/. F1's additional source and proof are under tracks/f1/cubic-divisor/. All constructed positives are identified as such; fixed empty cases are not success-rate estimates.

The most concrete new solver step

The original terminal equations retain a nonreduced symmetric algebra even when the scalar Dickson fibre is unramified. With S scalar roots, its scheme length is S^3 while its distinct unordered states number C(S+2,3).

The continuation instead imposes that the monic cubic h(T)=T^3-s1*T^2+s2*T-s3 divide D_S(T)-c, using a short repeated-squaring remainder circuit. For a separable fibre, the distinct-coordinate locus is reduced and has C(S,3) points. Separate two-equal and all-equal strata restore the repeated-coordinate solutions. Their total count is C(S+2,3).

This formulation completes the fixed depth-3 joint case that the earlier expanded formulation left unknown. At depth 3 its distinct membership algebra has dimension 56. At depth 4 the repeated strata complete but the distinct joint systems retain three recorded watchdog outcomes. Deep witnesses through depth 96 certify divisibility/Jacobians, not full relation solving. Neither a short circuit nor one completed case supplies a general cost bound.

Independent local checks and correction

Isolated reviewers checked the source and certificates without importing producer arithmetic, including separate point arithmetic, sparse polynomial division, norm factorization, rank calculations, lattice reconstruction and fresh isogeny/point-count computations. Their final coverage records are retained in reviews/. Blindness and custody deviations are recorded, including the unpublished-input exception; no official scientific review pass is inferred.

One precise F3 statement correction is retained in corrections/f3-coefficients-mod-r.md: the conditional uniform-target-coefficient lemma requires the aggregate coefficient vector to be nonzero in F_r^N. A nonzero integer vector c=(r,0,...) reduces

to zero and always yields B=0. This qualification does not change the implemented small-signed-coefficient search or its point certificates. It matters when extending the lemma to arbitrary multiplicities.

Other curves and the cover component

The P-224/Curve448 comparison uses generic D_S-c domains at S=16 and 32. P-224's split construction accepts canonical base sizes 8 and 18. Curve448's norm-one construction accepts sizes 2 and 4, after retaining 8 and 14 prime-subgroup rejections, and 6 and 14 non-lifts. All four domains and twelve fixed controls pass separate arithmetic implementations. Hasse-interval and prime-order witnesses certify group orders r and 4r. The same raw domain size therefore does not mean the same usable base size.

For the genus-three cover u^4=x^3-3x+b, the new derivation examines its two-dimensional Prym instead of treating the entire Jacobian as a new target. At p=3 mod4, Cartier kills dx/u^3 and x*dx/u^3. Frobenius anticommutation forces an even Prym polynomial T^4+A*T^2+p^2. The corrected Hasse--Witt congruence and Weil bound imply A=c*p for c=-2,-1,0,1,2. Exact P-256 residues exclude r from all five candidate Prym orders p^2+c*p+1. An isolated local rederivation confirms the target r-component is unique without computing the exact A. The full derivation and primary-source provenance are in analysis/cartier-proposal/; the isolated geometric review records its own proof and controls. No generic Jacobian solver or destination cost is supplied by this component argument.

Cardinality and complete-cost discipline

For a fixed signed base of 2M nonzero points, the number of represented nonzero d-summand targets is at most C(2M+d-1,d). Hence a necessary condition for half coverage of a uniform nonzero target is

2*C(2M+d-1,d) >= r-1.

The exact minimum-M boundaries are computed by integer bisection and retained in counting-screen.json. Approximate log2(M) values for arities 3 through 8 are 84.86,63.90,51.38,43.08,37.19,32.79. They are necessary capacities, not success lower bounds, attack-cost forecasts or claims about every possible algorithm. Target-dependent F3 bases require a separate argument.

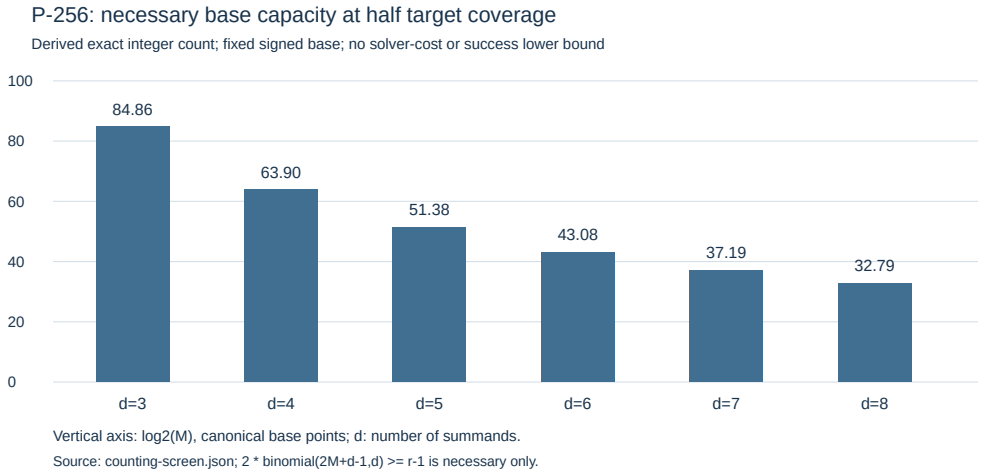

Figure 1: Derived fixed-base capacity

Every prospective comparison still needs construction, membership solving, root/sign/subgroup recovery, failed queries, partial-relation storage and stitching, independent rank, linear algebra where applicable, target attachment and memory. Cold setup and genuinely reusable setup are separate scenarios. Baselines receive the same subgroup, symmetry and setup reuse. The current software resource receipts are not matched ECDLP measurements.

Delivery and next concrete work

The local payload audit covers 716 changed research files and 11,009,714 bytes, with path/size/SHA-256 manifests and no credential-pattern matches. It is a delta atop the already public c35 snapshot; full third-party paper downloads and local rendering drafts are excluded. verify_delivery.py checks custody, parsing, explicit unknown outcomes and compact PDF bytes; it does not certify prescribed execution or mathematical promotion. The local ZIP is under /workspace/p256-research-delivery/. Publication remains subject to the pending explicit disclosure decision. Auto-merge is off.

The two next implementation questions are the F1 unexpanded cubic-remainder extractor and the F3 natural useful-dependency solver. The other tracks retain their named outstanding obligations, code and controls; none is silently closed. No whole ECDLP solve, uniform cost theorem, or practical speedup follows from this pass. Shared problem records track the exact missing interfaces.

Final local review outcome: the F1/F2 and F5/F7 joints passed within their stated coverage; F3/F4/F6/F8 certificates passed with the F3 coefficient qualification corrected additively. The twelve invalid inference/certificate controls in the latter panel were rejected. The geometric review independently reconstructs all four maps and the complementary point count, and checks a p=1 mod4 counterexample. These local outputs retain their custody, partial-blindness and unprobed-model deviations. They are not an official scientific promotion or approval to publish.